

Movie Rating Prediction using Machine Learning

Introduction

Movies play a significant role in entertainment and culture. Movie ratings influence audience decisions and commercial success. This project predicts movie ratings using machine learning based on genre, director, and actors.

Objective

To analyze movie data, preprocess it, perform EDA, and build a regression model to predict ratings.

Dataset

IMDb Movies India Dataset containing movie details such as genre, director, actors, and ratings.

Tools Used

Python, Jupyter Notebook, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn.

Data Preprocessing

Handled missing values, encoding issues, and categorical feature encoding using pipelines.

EDA

Histogram of ratings, genre distribution, and rating comparisons were analyzed.

Model

Linear Regression model using Pipeline and ColumnTransformer.

Evaluation

Evaluated using Mean Squared Error and R2 Score.

Conclusion

The model successfully predicts movie ratings and provides insight into factors influencing ratings.